July 13, 1959

MEMO TO: Manager, Department 516

SUBJECT: Two Papers to be presented at the Eastern Joint Computer Conference

Dr. John Cocke and I are each planning to give 20 minute papers at the EJCC which will be held in Boston, Mass., December 1-3, 1959.

Enclosed you will find the abstracts and summaries required for these papers. The papers will be based on a Product Planning Technical Report, Number P-17, dated June 12, 1959. A copy of this report is enclosed in order that you may have a more detailed account of the material to be covered in the papers. In particular, many of the figures in the report will be made into slides for the talks.

The deadline for the submission of the abstracts and summaries to the EJCC Program Committee is August 15, 1959, so we will appreciate your prompt clearance of these papers.

We feel that these talks will be very timely in view of the plans to announce STRETCH as a commercial system at about the time of the meeting.

H. G. Kolsky
Project Coordinator
Project 7000

HGK:jcj
Enclosures

cc: Mr. R. E. Merwin
Mr. D. W. Pendery
Mr. S. W. Dunwell
Dr. W. Buchholz
Dr. J. Cocke
Mr. E. Lester

July 1, 1959

ABSTRACT of paper to be given at the Eastern Joint Computer Conference, December 1 - 3, 1959

# The Virtual Memory in the Stretch Computer

John Cocke

International Business Machines, Research Lab.

Poughkeepsie, New York

The use of High Speed logical devices and more sophisticated design methods has resulted in an imbalance between the arithmetic and memory speeds. Various methods of "looking-ahead" have been proposed in order to permit the simultaneously operation of several blocks of memory and thus achieve a rate of operation compatible with the high arithmetic speeds. Further, machine speed may be increased by the overlapping of the bookkeeping operations with the principal computations. In order to achieve this a number of logical complexities must be overcome. The Virtual Memory is a device which satisfies the requirements of achieving the look-ahead and maintaining logical correctness in an asynchronous computer. This device will be described and the difficulties which it has overcome will be discussed.

July 1, 1959

Summary of paper to be given at the Eastern Joint Computer Conference, December 1 - 3, 1959/

# The Virtual Memory in the STRETCH Computer

John Cocke

International Business Machines, Research Laboratory

Poughkeepsie, New York

## I. INTRODUCTION

Early in the planning of the STRETCH computer it was seen that by using the latest solid state components in sophisticated circuits that it would be possible to increase the speed of floating point arithmetic by almost two orders of magnitude over that in existing computers. However, there seemed to be no possibility of developing on the same time-scale economically feasible large memories with more than a factor of ten or perhaps twenty increases in speed. As a result, the proposed system appeared to be in danger of being seriously memory-access limited.

Moreover, as the speed of the floating point operations increases, a larger and larger percentage of the computer's time is spent on "parasitic operations," i.e., operations whose only function is program control and data selection. It was obvious that a radically new machine organization was necessary in order to capitalize upon the possibilities opened up by the high arithmetic speeds in the presence of relatively slow memories.

At this time, the possibility of a "look-ahead" device was suggested in which an independent indexing arithmetic unit would prepare the effective addresses of instructions and initiate memory references to a multiplicity of memory boxes. The data thus fetched would be held in high speed buffer registers until needed by the arithmetic unit. This device would serve two desirable purposes; (1) some of the parasitic operations would be done in parallel and thus not delay the principal calculations, and (2) several memory boxes could be running simultaneously, giving the effect of higher memory speed.

## II. GENERAL DESCRIPTION OF THE SYSTEM

The major logically-independant blocks of the STRETCH computer are shown in Figure 1. Each of the units pictured may be considered as operating asynchronously. That is, each does its tasks as fast as possible independently of the others. In theory, each box could have its own clocking circuits and still operate properly. In practice, for economy's sake they are all timed by the same master oscillator, but this does not destroy their logical independence.

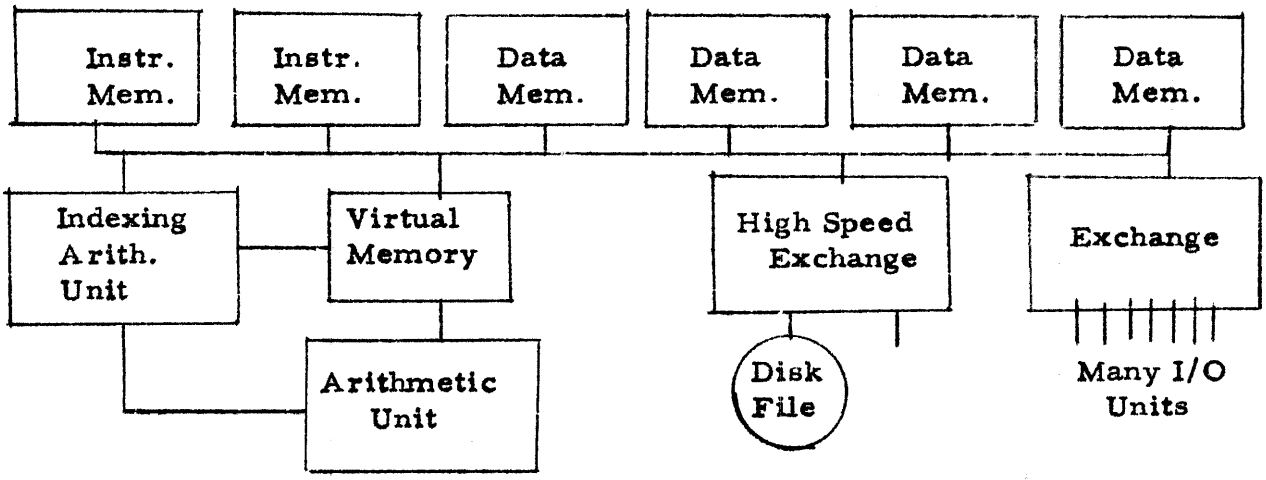
The bus control unit serves as a routing agent between the memories and the various data processing units. If two or more units make a request simultaneously the control unit assigns priorities in the following order: (1) High Speed Exchange, (2) Basic Exchange, (3) Virtual Memory, and (4) Indexing Arithmetic Unit.

The Indexing Arithmetic Unit fetches instructions, performs all necessary indexing operations and sends the instructions to be executed to the Virtual Memory.

The Virtual Memory fetches and receives the data required by the instruction and holds this data until the arithmetic unit is ready for it. The Virtual Memory also performs all store operations. It holds the data generated by the Arithmetic Unit or Indexing Arithmetic Unit until the memory to which the data must be sent is available. Thus the Virtual Memory acts not only as a "look-ahead" for instructions to be fed to the arithmetic unit, but also acts as a "look-behind" storage buffer.

FIGURE 1

SCHEMATIC OF SIGMA COMPUTER



## III. SOME LOGICAL PROBLEMS DUE TO OVERLAPPING

The actual design of such a "look-ahead" device posed a number of logical problems, particularly in connection with conditional branches.

With the above general organization in mind, let us discuss some of the logical problems posed by such a system. The first problem is a result of the very concept which enables us to obtain such great benefits from the stored program computer--the ability to treat instructions as data. In a system such as we have proposed there is a large amount of simultaneous operation. For example, the indexing arithmetic unit may be busy preparing an instruction before previous instructions have been completed, or even started by the arithmetic unit. One of these previous instructions may modify the instruction which is presently being indexed. The Virtual Memory must recognize this situation and allow the intervening instructions to be completed before doing the modified instruction.

A similar problem exists with respect to ordinary data. In order to operate several memories simultaneously, it is necessary to start obtaining data from these memories before the preceding operations have been completed. Yet, one of these operations may be a store into one of the data locations. The virtual memory must make provisions to insure that each instruction obtains the most up-to-date data as implied by the order of the program.

One of the novel features of the STRETCH computer is its elaborate interrupt system. Under this system whenever some unexpected occurrence arises, the program will be interrupted and control will pass to a special routine which is designed to take care of the case in question, then return control to the original program. In this situation the Virtual Memory must have provisions to retain enough information so that when an interrupt occurs we can resume the computation exactly where we left off. It must be able to recognize which of the changes that have been made in advance are not desired and should be obliterated and which are exact solutions that must be restored.

Another special case arises when a conditional branch on arithmetic results occurs. Here we will not know which of the two branches we should have taken until the prece ding instruction is executed. In the case the wrong path has been selected, the Virtual Memory must be prepared to drop the intermediate results which have been computed and pick up the correct branch in a way very similar to that of an interrupt.

Summing up all these logical problems, we may state that the fundamental rule for the Virtual Memory is that it must make the asynchronous and non-sequential computer give results identical to those which would be obtained by performing the program one instruction at a time in the order in which they are written.

## IV. GENERAL METHOD OF SOLVING LOGICAL PROBLEMS

In the STRETCH computer the index memory is addressable as ordinary memory so that index memory problems are reduced to the problems of ordinary memory. The only distinction being made is on the basis of whether the arithmetic unit or the indexing arithmetic unit desires the services of the memory cell. Each level of the Virtual Memory contains not only fields for buffering the data and operation codes, but also the location of data and operation codes plus control bits.

At any moment when data or instructions are desired from memory, a simultaneous comparison is made with all data addresses in the Virtual Memory. If a comparison occurs the general rule to be applied is that the Virtual Memory is substituting temporarily for that memory cell and the appropriate action taken. The details of this action will be described and it will be shown how this simple general principle (which motivated the name "Virtual Memory") guarantees the proper logical operation of the "look-ahead" system.

July 1, 1959

ABSTRACT of paper to be given at the Eastern Joint Computer Conference, December 1 - 3, 1959.

# Timing Simulation of the Stretch Computer

Harwood G. Kolsky
International Business Machines, Product Planning Dept.
Poughkeepsie, New York

In considering any complex machine organization which allows for the simultaneous operations of a number of memory units and more than one arithmetic unit, and includes such devices as the Virtual Memory, it is necessary to make a detailed timing simulation to evaluate the worth of this special hardware. The timing simulation must be carried out over a wide class of problems so that the computer design will not be biased toward one special purpose.

The general organization of such a simulation program for STRETCH written on the IBM 704 will be described. The results observed from varying such parameters as the number of memory units, interlace and allocation of memory addresses, number of Virtual Memory Levels, interference of input-output, arithmetic speeds, etc., will be discussed. Trade-off factors for such things as the speed of the indexing arithmetic unit, arithmetic unit, and memories will also be included.

July 1, 1959

SUMMARY OF PAPER to be given at the Eastern Joint Computer Conference. December 1 - 3, 1959

# Timing Simulation of the STRETCH Computer

Harwood G. Kolsky

International Business Machines, Product Planning Dept.

Poughkeepsie, New York

## A. General Considerations

During the logical design of STRETCH it was necessary to prove the value of the Virtual Memory concept and to assist in the selection of optimum values of various system design parameters. Examples of such parameters are: The number of memory boxes, interlace and allocation of memory addresses, and numbers of Virtual Memory levels. Also of interest were trade-off factors for speeds of indexing arithmetic unit, arithmetic unit, memories, etc.

In November 1957 the Timing Simulator (SIM-2) described here was written for the IBM 704. This program attempted to answer such questions quantitatively by simulating the time-wise operation of STRETCH on typical test programs coded in STRETCH language.

The basic logic of the 704 program follows the principles just described in the prece ding paper for the Virtual Memory. It should be stressed that the Simulator is a Timing Simulator and does not execute the instructions in an arithmetic sense. It traces the time-wise progress of the instructions through the components of the computer observing all the interlocks and time delays necessary for correct representation of the behavior of the machine.

One of the fundamental concepts in the STRETCH design is that of asynchronous operation of the components. This means that there are a large number of logical steps being executed at any one time in the computer, each of them proceeding at its own rate. To simulate this flow of many parallel continuous operations, we have broken the continuous time variable into finite time steps. The basic time step is taken as 0.1 microsecond in the Simulator.

By taking 0.1 microsecond as our quantum of time, we are automatically setting the scale of the smallest circuit entities which we will consider as being those which accomplish complete functions in a 0.1 microsecond or few multiples thereof. Thus, by using this philosophy, and considering many of the components of the computer as "black boxes", we greatly simplify the details which must be considered without introducing serious timing inaccuracies.

Our experience has indicated that more information was gained by making a large number of fast parameter studies using different configurations and programs than could have been obtained by a very slow, detailed simulation of a few runs with more precision per run. Even so our time scale is too fine to make serious Input-Output application studies. These would require a simpler Simulator having at least a factor of 10 coarser basic time interval.

## B. Logic of the Simulator

In the asynchronous organization of STRETCH there can be many major components operating at any one time. To achieve this parallel effect in the Simulator we essentially "hold time still" and scan the entire machine representation at each time step. Although every major block of the program is traversed at each time step, if there is no activity required in a given block, only a few tests need be made by the code.

If in this process it is determined that a given logical unit should do an operation, the time interval required for the operation is obtained from a table of constants. The speed of the various logical units can thus be changed parametrically by changing the values in the tables. A constant obtained from the tables is inserted into a memory location called the time counter for that unit. At each time step the program reduces this counter by one until it reaches zero. Thus, the fact that the counter is non-zero can be used to indicate that the particular logical unit is busy and not available to service other requests. When the counter is zero the unit can consider a new input.

In addition to the time counters many of the logical blocks contain other conditions or interlocks which effect the operation of the block. These conditions are stored in the program and tested before action is undertaken.

It is interesting to note that since the Simulator simulates timing only, not the arithmetic or indexing functions, the sequence of instructions to be executed must be furnished as a "string" with all loops unwound. However, to make the computer behave as it actually would, the loops must be furnished with "wrong way" paths given for the cases where the computer

would take such paths. Also one must furnish more than enough information along such paths since it is difficult to predict in advance how far the computer will get down the wrong path before it is called back.

Parameters are changed from one run to another by use of control cards. The control cards are set up in such a way that any number of parameters may be changed between runs. Results are given either as detailed timing charts or as summary listings for each problem. The usual procedure has been to print only summary results while making a series of parameter studies. The detailed timing charts as printed on the 704 for most problems would be about 50 feet long for each run. Since over a 1000 cases have been run, it is clear that only a few cases could be printed in full detail.

## C. Results of Parameter Studies

When the Simulator Program was completed, we undertook a series of studies in which the main parameters describing the STRETCH system were varied one or two at a time in order to get a measure for the importance of different effects. After this we began to specialize the studies towards answering specific questions in the STRETCH design.

Tables and curves showing the results of these parameter studies along with a discussion of some of the general properties discovered will be presented.

IBM

Date: July 30, 1959
From: H. G. Kolsky
Department: 539
Location: Poughkeepsie

To: Mr. Paul Nelson
Department: 516
Location: Poughkeepsie

CC: Mr. R. E. Merwin
Mr. D. W. Pendery
Mr. S. W. Dunwell
Dr. W. Buchholz
Dr. J. Cocke
Mr. E. Lester

Subject and/or Reference: Two Papers to be Presented at the Eastern Joint Computer Conference

Memo same subject dated 7/13/59

In the referenced memo, I sent your department copies of abstracts and summaries of papers which Dr. Cocke and I plan to give at the EJCC in December. The deadline for the submission of the abstracts and summaries is August 15, 1959.

Dr. Cocke is leaving for vacation in South America from August 8 to August 24. I have recently been transferred to the Federal Systems Division and will be in the process of moving for the next few weeks.

I have written letters for transmitting the abstracts and summaries to the program committee on August 14 and am leaving them with Dr. Werner Buchholz. He will be glad to answer my questions about them during our absence.

Again I emphasize the importance to the Company of having these STRETCH papers on the program of the EJCC.

H. G. Kolsky

HGK:jcj

August 14, 1959

Dr. J. H. Felker, Chairman
EJCC Program Committee
Bell Telephone Laboratories, Room 5C-101
Mountain Avenue
Murray Hill, New Jersey

Dear Sir:

I am happy to forward for your consideration the abstracts and summaries of two papers for the Eastern Joint Computer Conference in Boston, December 1-3, 1959.

As you can see, both of the papers deal with the IBM STRETCH computer. They form a pair of complementary papers with Dr. Cocke's being first.

If you have any questions concerning the papers in the next few weeks, please direct them to Dr. Werner Buchholz, IBM South Road Laboratory, Poughkeepsie, New York.

Very truly yours,

GL-4-1000

H. G. Kolsky

HGK:pen

cc: Dr. W. Buchholz
Dr. J. Cocke

bcc; Messrs. S. W. Dunwell
E. Lester
R. E. Merwin
D. W. Pendery

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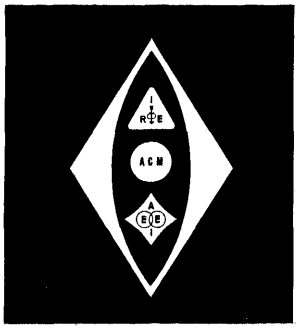
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**1959**

**Eastern Joint**

**Computer Conference**

**December 1, 2, 3, 1959**

**Statler Hilton Hotel, Boston**

In Reply Address
Bell Telephone Laboratories
Mountain Avenue
Murray Hill, New Jersey
Room 5C-101

August 20, 1959

MR. H. G. KOLSKY
International Business Machines Corp.
Data Processing Division
Box 390
Poughkeepsie, New York

Dear Mr. Kolsky:

Thank you for your letter of August 14, 1959, sending the abstract and summary of two papers by John Cocke and yourself entitled "The Virtual Memory in the Stretch Computer" and "Timing Simulation of the Stretch Computer". We will be in touch with you in September after the Program Committee has reviewed all the papers proposed for the EJCC.

Very truly yours,

R A Kudlich

R. A. KUDLICH, Vice-Chairman
1959 EJCC Program Committee

RAK/gn

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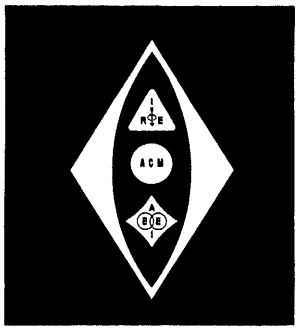
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**1959**

**Eastern Joint**

**Computer Conference**

**December 1, 2, 3, 1959**

**Statler Hilton Hotel, Boston**

In Reply Address
American Telephone &
Telegraph Company
195 Broadway
New York 7, New York
Room 1122

9-21-59 1 1/c

September 21, 1959

Messrs. H. G. Kolsky
J. Cocke
International Business Machines Corp.
Product Development Laboratory
P.O. Box 390
Poughkeepsie, New York

Gentlemen:

It is a pleasure to inform you that your Paper "The Virtual Memory in the Stretch Computer" has been accepted for presentation at the 1959 EJCC. I believe that J. Haanstra and R. A. Kudlich have discussed with you our desire to have the two papers on the Virtual Memory combined into one.

You will be interested to know that four out of every five Papers submitted for the Conference had to be rejected because of limited program time. The fact that your summary was accepted testifies to the expectations that we have for your Paper. I hope you recognize your responsibility for making an interesting, intelligible and well rehearsed presentation at the Conference. Since you will have only twenty minutes to present your talk, I suggest that you choose a few high points to cover in the oral presentation. Please bear in mind that the specialists in your subject, interested in detail, will have ample opportunity to study your full Paper in the proceedings of the Conference.

To accommodate the audience expected, it may be necessary to show a duplicate set of your slides in a room adjacent to the main hall. Please bring two sets of slides with you to the conference. Experienced speakers find that the requirements for a good slide are quite different from those for a manuscript illustration. May I request that you give special attention to preparing simple slides and to limiting their number so that the audience will have adequate time to study each one.

We will need a new hundred word abstract by October 5th for inclusion in the printed program.

Felker
Murray Hill, N.J.
201- CR-3-6000
Ext. 2944

Acceptance of your Paper, and the right to present it at the Conference, is conditional upon four copies of the completed Paper being received by H. Anderson on or before November 15, 1959. No exceptions will be made to this rule. Instructions for the preparation of your manuscript are enclosed. It is important that these instructions be followed. When you send your manuscript, please include information on the size of the slides you will use and on any other visual aids you may wish to employ. At the same time, please enclose a one-hundred word biography of whichever author will make the oral presentation.

Registration fees will be waived for the speakers and each speaker will receive a complimentary ticket to the award dinner, which will be held on the evening of December 3. It is important that all speakers attend because at this dinner the award of $300 will be made for the best presentation at the Conference of a Paper describing significant work in the computer field. The award was set up in recognition of the fact that careless and obtuse presentations of technical papers are frequently a disgrace to the speaker and an anesthetic to the audience. I urge you to arrange for at least one rehearsal of your presentation before your associates and I recommend the use of a tape recorder at the rehearsal as an aid to a fruitful post mortem.

Sincerely,

J. H. Felker

Attachment
Manuscript Instructions

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**1959**

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**December 1, 2, 3, 1959**

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In Reply Address
P. O. Box 5,
Needham 94, Massachusetts

## MANUSCRIPT INSTRUCTIONS TO AUTHORS

You will have two means of cummunicating your 1959 Eastern Joint Computer Conference paper to the computer field, orally and written.

Past conferences have shown that the Proceedings of the E.J.C.C. are an important and lasting record of the meeting. To maximize the usefulness of this material two important points must be kept in mind.

1. the written record must be distributed soon after the conference.

and 2. the material must be prepared in a clear manner.

## DEADLINES

Four copies of your final manuscript and all drawings and photographs must be in the hands of the Publications Committee by November 15, 1959. Send them to:

Harlan E. Anderson
EJCC Publication Committee
Digital Equipment Corporation
Maynard, Massachusetts

E.J.C.C. Steering Committee voted not to print papers that failed to meet this deadline.

During the conference itself (December 1-3) you will be contacted by a member of the Publications Committee for a final check on any editorial changes that may have been made on your paper. It is hoped that galley proofs from the printers will be available at that time.

Review of the text of questions and answers as recorded by stenotypists will also be carried out during the conference if at all possible.

## MATERIAL PREPARATION

### Title

The title should be short if possible but not so short that the subject will not be indicated clearly. The author should remember that a paper is indexed by significant words in the title and that many readers of the PROCEEDINGS select the papers they read from the table of contents. The title, therefore, should be carefully chosen. Six to eight words is the usual maximum length.

### Abstract

The abstract should be really informative and should outline the essentials of the paper within the compass of about 200 words or less. It should be meaty enough to serve as a complete abstract for such services as Science Abstracts or the Abstracts and References section of the PROCEEDINGS. The author should ask himself if he is willing to accept his abstract, without the rest of the paper, as a fair statement of what he has accomplished. The problem should be clearly stated, the method of attack outlined, and numerical results, where possible, included. In general neither references nor equations should appear in the summary. The essence of a good abstract is a concise statement in plain English of just what has been done and why.

### Figures and Tables

It is very helpful to readers of all classes if tables, figures, curves, etc. can be arranged so that they are self-explanatory and can be used with a minimum of reference to the text. The quantities and units used in plotted curves should be given clearly, and the captions worded to convey as much information as possible to the engineer who is merely leafing through his copy of the PROCEEDINGS. If his attention is caught by curves that mean something to him, he is much more likely to read the paper. A caption such as "Plot of $X_0/y$ versus s for various values of the parameter n," will not enlist the interest of many readers. It is better to say, "Axial sound-pressure curves of electrostatic loudspeaker for various diaphragm thicknesses."

### Manuscript

Manuscripts should be typed on one side of the sheet only and double spaced. All illustrations should be referred to in the text by figure number. All footnotes should be numbered.

Footnote references must be complete. Authors' initials, title of article, volume and page numbers, and month of publication should be included. The complete reference should be as follows:

For a periodical: R.N. Hall "Power rectifiers and transistors." PROC. I.R.E., vol. 40, pp. 1512-1518; November, 1952.

For a book: W. A. Edson, "Vacuum Tube Oscillators," John Wiley and Sons, Inc., New York, N.Y., pp. 170-171; 1948.

Illustrations

Drawings for printing must be in black ink on white paper or on tracing cloth. Photographs must be glossy prints. The size should not exceed 8½ by 11 inches.

It should be borne in mind that most illustrations will be reduced in size to a 3½ column width when printed. It is particularly important, therefore, that all letters, numbers, and lines be drawn large enough and heavy enough to remain legible after reduction.

Drawings with typewriting on them are not acceptable. All information to be reproduced must be hand lettered in ink.

Graphs should be drawn with only the major coordinate lines showing. A chart containing a large number of closely spaced line will not reproduce legibly.

Captions, if they are included on the drawing itself, should not appear within the area to be reproduced. They should be placed under the illustrations.

Please feel free to contact me, if you have questions concerning any of the material above.

Harlan E. Anderson
Chairman, Publications Committee

POUGHKEEPSIE

POUGHKEEPSIE

Dept. 516
Kenyon 224

October 1, 1959

MEMO TO: Messrs. Erich Bloch
John Cocke
R. J. Evey
L. Hellerman
H. G. Kolsky

SUBJECT: Outside papers accepted for Eastern Joint Computer Conference.

In the manuscript instructions to authors received from the EJCC Program Committee, they specify quite forcibly that the complete papers must be in their hands by November 15. This does not allow much time for preparing, reviewing, and clearing the papers within IBM.

We urge you to make every effort to get drafts of your full papers into our hands by October 16, if possible. This will give us four working weeks to complete the job, which is reasonable but certainly not leisurely. Please keep in mind that some of these papers have patent implications that will take some time to resolve; also, that we will need independent reviews by qualified persons outside your own areas. Concerning the latter, we will appreciate your suggestions.

If we in the Publications Department can help you in the early stages of planning text, art work, etc., please get in touch with us.

E. F. Boomhower
P. J. Nelson

E. F. Boomhower
P. J. Nelson
Publications Department
Engineering Communications

EFB/PJN:pam

cc: Mr. W. Buchholz
Mr. S. W. Dunwell
Mr. J. F. Hanifin
Mr. J. P. Lazarus
Mr. H. T. Marcy
Mr. C. A. Merritt

P. J. Nelson
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Whitlock Associates

**1959**

**Eastern Joint**

**Computer Conference**

In Reply Address
P. O. Box 5,
Needham 94, Massachusetts

OCT 29 1959

**December 1, 2, 3, 1959** October 21, 1959

**Statler Hilton Hotel, Boston**

Mr. H. G. Kolsky
International Business Machines Corp.
Product Development Laboratory
P.O. Box 390
Poughkeepsie, New York

**Dear Sir:**

**You are listed to deliver the following paper at the Eastern Joint Computer Conference in Boston next December.**

**Title:** "The Virtual Memory in the Stretch Computer"

**Time and Date:** 9:00-12:00 a.m. Wednesday, December 2.

**Because interest on the part of the technical and popular press is at an unprecedented level, we are expecting more media representatives will attend the Conference than ever before. In order to provide the best service possible, and to assure accurate reporting, we need advance copies of a summary of your paper so the press may be provided access to it prior to your presentation. As you know, it is not possible for a writer to do justice to a paper if he must write his story from notes made while attending the session. With advance copies available, reporters will not only write more accurately, but will also be able to seek out authors for interpretation or amplification well ahead of deadlines.**

**This is to request that you forward twenty five (25) copies of a 1000 word summary of your paper so that they will reach here not later than Friday, November 20. Copies should be mailed to:**

**George D. Wood, Jr.**
**Office of Public Relations, Room 3-339**
**Massachusetts Institute of Technology**
**Cambridge 39, Mass.**

**We shall also contact the public relations office of your firm to let them know of your paper, and to seek their cooperation on preparing news releases for popular media.**

**Your cooperation will be most appreciated.**

**Sincerely,**

**Publicity and Printing Committee**

*Chairman*
Frank E. Heart,
Lincoln Lab.

*Program Committee*
Jean H. Felker, Bell Labs.

*Publication Committee*
Harlan F. Anderson,
Digital Equipment Corp.

*Local Arrangements Committee*
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Philip R. Bagley,
Lincoln Lab.

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David L. Bailey, MITRE
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National Co.

*Exhibits*
Howard I. Cohen, Sylvania

*Exhibits Management*
John Leslie
Whitlock Associates

**1959**

**Eastern Joint**

**Computer Conference**

In Reply Address
American Telephone &
Telegraph Company
195 Broadway
New York 7, New York
Room 1122

**December 1, 2, 3, 1959**

**Statler Hilton Hotel, Boston**

October 26, 1959

MR. H. G. KOLSKY
International Business Machines Corp.
Federal Systems Division
Omaha Engineering Office
3104 Farnam Street
Omaha 31, Nebraska

Dear Mr. Kolsky:

Thank you for your letter of October 5, 1959, sending us the new abstract for the Paper entitled "The Virtual Memory in the Stretch Computer" by John Cocke and yourself. The revised abstract will appear in both the final Conference Program and in the Conference Proceedings.

Very truly yours,

R. A. KUDLICH, Vice Chairman
1959 EJCC Program Committee

OCT 29 1959

**Abstract of Paper to be given at the Eastern Joint Computer Conference, December 1 - 3, 1959**

# The Virtual Memory in the Stretch Computer

John Cocke
and
Harwood G. Kolsky

International Business Machines Corporation
Poughkeepsie, New York

The use of high speed logical devices and more sophisticated design methods has resulted in an imbalance between the arithmetic and memory speeds. The Virtual Memory provides a method of "looking-ahead" in order to permit the simultaneous operation of several blocks of memory in an asynchronous computer while maintaining logical correctness. Machine speed is further increased by overlapping the bookkeeping operations with the principal computations. To measure the effectiveness of the Virtual Memory, a timing simulation program for STRETCH was written on the IBM 704. This program enables us to study a wide variety of parameters for design optimization.

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John Leslie,
Whitlock Associates

**1959**
**Eastern Joint**
**Computer Conference**

**December 1, 2, 3, 1959**
**Statler Hilton Hotel, Boston**

In Reply Address
P. O. Box 5,
Needham 94, Massachusetts

In Reply Use
Address Below

NOV 9 1959

November 3, 1959

Messrs. H.G. Kolsky
J. Cocke
International Business Machines Corp.
Product Development Laboratory
P. O. Box 390
Poughkeepsie, New York

Dear Messrs. Kolsky and Cocke:

I would like to remind you of the approaching deadline for the manuscript of your paper for the 1959 Eastern Joint Computer Conference.

Four complete copies of your manuscript including all drawings and charts should be in my hands by November 15. The EJCC steering committee voted that papers not meeting this deadline would neither be printed in the Proceedings of the conference nor would the speakers be permitted to deliver their paper at the conference.

As you probably know, a concentrated effort is being made to print the proceedings promptly. The Publication Committee is planning to check with you during the conference on any editorial changes in your manuscript and/or question and answer transcripts. Would you or a representative of your paper please report to the Publication Committee Room in the Hotel Statler to arrange for a time to take care of this? In the case of 1st and 2nd day papers, the day after your paper would be an appropriate time. Authors of all other papers try to check in on the first day of the conference concerning editorial changes to manuscripts and make plans for reviewing questions and answers.

I hope that you will find your participation in the 1959 EJCC a rewarding experience.

Sincerely,

Harlan E. Anderson

Harlan E. Anderson, Chairman
1959 EJCC Publication Committee
Digital Equipment Corporation
Maynard, Massachusetts

*Chairman*
Frank E. Heart,
Lincoln Lab.

*Program Committee*
Jean H. Felker, Bell Labs.

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**1959**

**Eastern Joint**

**Computer Conference**

**December 1, 2, 3, 1959**

**Statler Hilton Hotel, Boston**

In Reply Address
xP.O.Box 73x
xLexington 73, Massachusettsx

Mass. Inst. of Tech.
Lincoln Laboratory
P. O. Box 73
Lexington 73, Mass.

NOV 5 1959

4 November 1959

Messrs. Harwood G. Kolsky and John Cocke
International Business Machines Corporation
Product Development Laboratory
P. O. Box 390
Poughkeepsie, New York

Dear Sirs:

The Management of the 1959 Eastern Joint Computer Conference would like to offer a complimentary registration for the conference to all authors of papers on the technical program. Registration material may be picked up at the advance registration desk of the conference. These registrations will include a complimentary copy of the Proceedings, when issued.

In addition, the Conference Management would like to invite the actual individual speaker to accept a complimentary banquet ticket and a place at the speakers' table for the banquet on Thursday, December 3. Since I do not yet know which of you will be delivering this paper, I would appreciate this information as soon as possible so that proper arrangements may be completed.

Sincerely yours,

Frank E. Heart

Frank E. Heart
Chairman

FEH:bic

INTERNATIONAL BUSINESS MACHINES CORPORATION
FEDERAL SYSTEMS DIVISION
SYSTEMS ENGINEERING OFFICE
3104 FARNAM STREET
OMAHA 31, NEBRASKA

November 13, 1959

Dr. Harlan E. Andersen
EJCC Publication Committee
Digital Equipment Corporation
Maynard, Massachusetts

Dear Dr. Andersen:

In accordance with the instructions contained in Dr. Felker's letter of September 21, 1959, I am enclosing four copies of the manuscript entitled, "The Virtual Memory in the STRETCH Computer", by John Cocke and Harwood G. Kolsky for inclusion in the Proceedings of the E. J. C. C. December 1 - 3, 1959.

Also enclosed is a one-hundred word biography of the speaker. For visual aids I plan to use 35 mm slides.

Sincerely,

H. G. KOLSKY

HGK/jlv
Enclosures

Abstract of Paper to be given at the Eastern Joint Computer Conference, December 1 - 3, 1959

# The Virtual Memory in the Stretch Computer

John Cocke
and
Harwood G. Kolsky

International Business Machines Corporation
Poughkeepsie, New York

The use of high speed logical devices and more sophisticated design methods has resulted in an imbalance between the arithmetic and memory speeds. The Virtual Memory provides a method of "looking-ahead" in order to permit the simultaneous operation of several blocks of memory in an asynchronous computer while maintaining logical correctness. Machine speed is further increased by overlapping the bookkeeping operations with the principal computations. To measure the effectiveness of the Virtual Memory, a timing simulation program for STRETCH was written on the IBM 704. This program enables us to study a wide variety of parameters for design optimization.

## Biography of Dr. Harwood G. Kolsky

Dr. Kolsky obtained his bachelor's degree from the University of Kansas and his doctorate in Physics from Harvard University. During World War II he did work in cryptography and communications in the U. S. Signal Corps. He spent 7 years on the staff of the Los Alamos Scientific Laboratory in the field of digital computer applications as an Associate Group Leader in the Theoretical Division. He has at various times held part-time teaching positions at Kansas, Harvard, and New Mexico Universities.

In 1957 he joined IBM as a member of the STRETCH Project Product Planning group. He was recently made Assistant Manager of IBM's SAC Intelligence Project in Omaha, Nebraska.

Date: November 3, 1959

From: Dr. H. G. Kolsky, Systems Engineering Office

Subject and/or Reference: Paper for Eastern Joint Computer Conference

To: Mr. P. J. Nelson, Publications Department
Engineering Communications, IBM Poughkeepsie

Re: My memos, this subject, June 13, '59 and July 30, '59
Your memo October 1, '59
(Also Ref Correspondence with J. H. Felker and R. A. Kudich)

Enclosed are the vellum masters of the manuscript, "The Virtual Memory in the STRETCH Computer" by John Cocke and Harwood G. Kolsky. This represents a combination of the two papers originally proposed for the EJCC. They have been combined as a result of Dr. Felker's request.

Except for minor editing changes, the manuscript is a direct copy from Product Planning Technical Report No. P-17 on the same subject dated June 12, 1959. We have intentionally removed most of the tables of quantitative results and any other details which might be misconstrued as embarrassing to the Company.

The verbal presentation will be considerably shorter than this printed version. In particular, I intend to omit Section III entirely in the verbal presentation and to replace it with a few general remarks.

According to the directions, which we have received from Dr. Felker, four copies of the above manuscript should be in the hands of the Publications Committee of the EJCC by November 14, 1959.

I am also enclosing a copy of my biography and a letter of transmittal as called for in Dr. Felker's instructions.

Because of my present duties and location I have found it very difficult to make the required copies and coordinate the many last-minute details required to get the paper "off the ground". I am therefore calling on you and your Department for the help you have so generously offered to take care of such details. I am having the slides made here.

You may call me at WEbster 9200, Omaha, Nebraska, or preferably you may get in touch with Dr. J. Cocke in Poughkeepsie for questions.

Subject: Paper for Eastern Joint Computer Conference
To: Mr. P. J. Nelson, IBM Poughkeepsie November 3, 1959

Please remember that you now have the only master copy in existence. I would appreciate your sending me a few ozalid copies of the paper at your convenience. Thank you very much.

H. G. KOLSKY
Assistant Manager
438-L Project, Omaha

HGK/jiv

Encl:
1 "Virtual Memory" MS Ozalid Master
1 Set of Figures for MS
1 Biography ozalid master
1 Abstract of Virtual Memory Paper
1 Letter of Transmittal

cc S. W. Dunwell, Poughkeepsie
R. E. Merwin, Poughkeepsie
E. Lester, Poughkeepsie

*Chairman*
Frank E. Heart,
Lincoln Lab.

*Program Committee*
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**1959**
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**December 1, 2, 3, 1959**

**Statler Hilton Hotel, Boston**

In Reply Address
American Telephone &
Telegraph Company
195 Broadway
New York 7, New York
Room 1122

November 18, 1959

Mr. H. G. Kolsky
1133 South 94 Street
Omaha, Nebraska

Dear Mr. Kolsky:

This is to request that you meet with your Session Chairman, Mr. Morris Rubinoff, in Room 435 on December 2 at 8:30 a.m. to discuss final arrangements for the presentation of your paper.

I believe that attendance at this meeting is essential to insure that there is no misunderstanding about time warnings, slide projection arrangements, etc.

Sincerely,

J. H. Felker

November 19, 1959

Dr. Harlan E. Anderson
EJCC Publication Committee
Digital Equipment Corporation
Maynard, Massachusetts

Subject: "The Virtual Memory in the Stretch Computer",
by John Cocke and Harwood G. Kolsky

Dear Dr. Anderson:

Enclosed are copies of two corrected pages from the above manuscript, involving a technical inconsistency in the text (p. 20) and a short section of acknowledgement (p. 35). These I spoke about when I delivered the paper to your office on Nov. 16.

Would you please replace the original pages with these in your copies of the manuscript, particularly the copy that is to be sent to the printer? If you can arrange to do this, the authors would appreciate it.

If there is any other material that we can furnish, please let us know. Thank you for your cooperation.

Sincerely,

P. J. Nelson
Editor, Publications Dept.

PJN:am
Enc.
cc: Mr. John Cocke
Mr. Harwood G. Kolsky
Mr. S. W. Dunwell

November 25, 1959
H. G. Kolsky, Assistant Manager
Systems Engineering Office, Omaha

EJCC Paper

P. J. Nelson
Editor, Publications Dept.

I wish to express Dr. Cocke's and my appreciation for your efforts in preparing our paper, "The Virtual Memory in the STRETCH Computer", for delivery to the EJCC Publication Committee. The work which you did to get the figures and tables redrawn in such a short time is particularly appreciated. It is really gratifying for an author to have such support from a service department.

H. G. KOLSKY
Assistant Manager
438L Project, Omaha

HGK/jlv

cc Dr. John Cocke - Poughkeepsie
Mr. S. W. Dunwell - Poughkeepsie
Mr. R. E. Merwin - Poughkeepsie